

Graph-based Prompt Learning with Mixture of Experts for Multi-task Corporate Profiling

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Corporate profiling serves as a critical analytical tool for modern enterprises, enabling data-driven decision-making in investment strategies, risk assessment, and strategic planning. It requires integrating quantitative metrics, qualitative insights, and network relationships to capture a company's role in the business ecosystem. However, traditional methods struggle to synthesize heterogeneous data and model complex interdependencies among corporations, news, and market dynamics, often addressing these aspects in isolation. To address these challenges, this article introduces Financial Graph-based Mixture of Experts Prompt Learning (FGMPL), an innovative framework that unifies graph prompt learning with a multi-task paradigm for corporate profile modeling. The proposed framework reformulates node- and edge-level tasks into a coherent graph-level representation and employs multi-view contrastive learning to effectively integrate textual details with relational structures. Moreover, a novel Financial Multi-Experts Prompting mechanism—with learnable tokens coupled with a Mixture of Experts (MoE) design—is presented to enhance the processing of heterogeneous

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graph data and bridge the gap between pre-training and downstream tasks. To further improve adaptability, a meta-learning-based prompt tuning strategy is incorporated, enabling rapid transition to various downstream applications. Extensive experiments on real-world financial graphs show that FGMPPL consistently outperforms strong pre-training and graph-prompting baselines across corporate performance prediction, relationship prediction, and news classification in both full-data and few-shot settings. In addition, cross-market transfer on a NASDAQ dataset and interpretability/efficiency analyses further demonstrate its robustness and practical applicability.

CCS Concepts: • **Information systems** → **Data mining**; • **Computing methodologies** → *Learning latent representations*; • **Networks** → *Network algorithms*;

Additional Key Words and Phrases: Graph Neural Networks, Prompt Learning, Mixture of Experts, Meta-learning, Corporate profiling

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1 Introduction

In the contemporary business environment, constructing comprehensive corporate profiles is essential for guiding decision-making among stakeholders—including investors, regulators, and business partners [48, 95]. Such profiles offer critical insights into various dimensions of business entities, encompassing fundamentals, operations, products, market presence, and competitive positioning. Modern methodologies transcend conventional financial metrics by integrating quantitative data, such as financial statements and market performance, with qualitative insights derived from news reports, regulatory filings, and analyses of corporate networks [34, 96]. This integration fosters a multi-faceted information ecosystem wherein the heterogeneous and interdependent nature of the data imposes substantial challenges for analysis.

The construction of comprehensive company profiles generally encompasses three primary tasks: (1) predicting company performance—forecasting financial indicators such as stock prices and revenue trends [10, 46, 99]; (2) analyzing inter-company relationships—examining interactions like competition, collaboration and supply chain dynamics [2, 56]; and (3) classifying financial news—organizing and interpreting unstructured textual data to assess its impact on corporate reputation, investor sentiment, and market behavior [14, 38, 64]. These three tasks collectively capture different aspects of company profiles, providing a multi-dimensional perspective on corporate operations and interactions. Specifically, in the domain of company performance prediction, methodologies have evolved from traditional market indicator analyses [7, 11, 21] to knowledge graph-based approaches that integrate event-driven [22] or relationship-driven [98] insights, thereby enhancing predictive accuracy. Secondly, regarding inter-company relationships, the methodological spectrum spans text mining [39, 55], similarity-based methods [52, 67], probabilistic models [17, 20], and learning-based approaches [5, 29, 58]. Recent advances in **Graph Neural Networks (GNNs)** have been noteworthy; for instance, RGCN employs attention mechanisms to model both cooperative and competitive enterprise relationships [19], while SPGCN decomposes complementary and substitutive interactions [18], thereby facilitating multi-dimensional feature analysis for investment strategies. Thirdly, from the perspective of financial news classification, deep neural networks have enhanced corporate disclosure text interpretation through context-aware reinforcement [44]. And

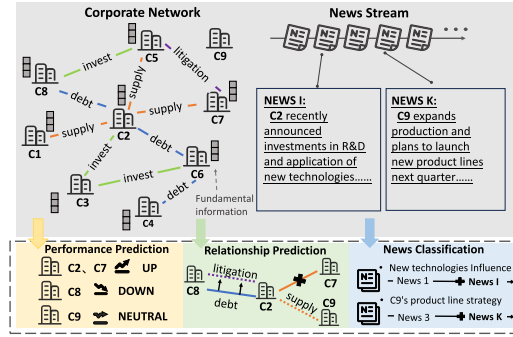


Fig. 1. A unified framework that integrates three traditionally isolated corporate analysis dimensions.

the SePaL framework leverages event graphs and self-supervised learning to generate dynamic enterprise representations [96], offering a solution for comprehensive company profile analysis. Collectively, these tasks establish an integrative feedback loop that synthesizes diverse insights to foster a comprehensive understanding of corporate profiles [61, 84].

However, several critical challenges persist. First, the continuous emergence of new nodes and the inherent diversity of structured data impede the adaptation and generalization of conventional end-to-end learning approaches. As illustrated in Figure 1, the complexity of corporate networks is exacerbated by emerging news events, which dynamically alter the network structure. Furthermore, the application of pre-training techniques to financial ecosystems often suffers from a misalignment between pre-training tasks and downstream objectives, due to the unique characteristics of these domains. Second, the heterogeneous graph data found in financial ecosystems—which consists of multiple types of nodes and relationships—presents additional difficulties in extracting the relevant information for downstream tasks. Approaches that isolate these elements and focus solely on individual aspects risk overlooking the inherent interconnections within the system, potentially leading to biased predictions. For example, focusing solely on performance metrics might mistakenly predict C7’s growth because of network spillover effects, while neglecting adverse influences arising from complex supply chain relationships. Similarly, analyzing news events in isolation might mistakenly categorize News I and News K as unrelated, despite the possibility that they are linked through underlying enterprise relationships. Finally, comprehensive corporate profiling inherently involves multiple, interrelated tasks. Thus, multi-task learning frameworks are essential to manage these interdependencies effectively. Such frameworks are designed to leverage shared information across tasks, thereby enhancing overall model performance—particularly in scenarios with sparse data or frequent emergence of new entities and events, which is common in the financial domain.

To address these challenges, we propose a unified framework, **Financial Graph-based Mixture of Experts Prompt Learning (FGMPL)**, to construct comprehensive corporate profiles that reflect the multi-faceted and evolving nature of companies in the financial ecosystem. Concretely, we formulate three key aspects of corporate profiles—performance metrics, business relationships, and news coverage—into a unified learning paradigm by extracting a local induced subgraph centered at each target node/edge and using it as a graph-level instance. By integrating these dimensions, FGMPL captures the essential attributes that define a company’s characteristics and context, thereby enabling stakeholders to gain a holistic and dynamic understanding of corporate entities. Initially, our approach employs multi-view contrastive pre-training, effectively integrating

structural information from news texts and corporate relationship networks to construct multi-dimensional representations. Subsequently, we adopt a prompt tuning strategy in which specifically designed prompts reformulate downstream tasks to align more closely with pre-training objectives, enhancing the effectiveness of knowledge transfer. Notably, through the graph prompt learning paradigm, FGMPL flexibly accommodates newly introduced nodes without requiring full model retraining, effectively addressing the challenges posed by the continuous emergence of new entities in the financial ecosystem. To deal with heterogeneous information and complex relationships, we propose a Financial Multi-Experts Prompting mechanism, combining prompt-based learning with sparse top-k **Mixture of Experts (MoE)** routing to unify upstream–downstream learning and yield task-dependent expert specialization for better interpretability and adaptation to new nodes. Furthermore, meta-learning techniques are incorporated into the prompt tuning process to facilitate rapid adaptation to diverse downstream tasks while mitigating the risk of negative transfer. Finally, we validate the proposed FGMPL framework on multiple tasks—including corporate performance prediction, relationship prediction, and news classification—demonstrating its efficacy in comprehensively analyzing complex interactions within the financial ecosystem. To evaluate cross-market generalization, we additionally construct a NASDAQ (Q2 2016) heterogeneous news–corporate graph by integrating multiple public sources and conduct transfer experiments from the Chinese market to the U.S. market. FGMPL consistently outperforms strong baselines in three tasks. The contributions of this article can be summarized as follows:

- To the best of our knowledge, we are the first to systematically define a comprehensive and practical problem for corporate profile construction, incorporating three key tasks—performance prediction, relationship analysis, and news classification—and propose a solution framework, FGMPL, that effectively integrates these tasks into a unified learning paradigm capable of handling both node-level and edge-level tasks simultaneously.
- We propose an innovative Financial Multi-Experts Prompting mechanism, which uniquely integrates prompt templates with a MoE architecture, enabling specialized experts to collaboratively process distinct aspects of heterogeneous financial information and effectively address the complexities of financial data, akin to how domain experts analyze different components of the financial ecosystem.
- We incorporate meta-learning techniques into our graph prompting framework to facilitate efficient adaptation to sparse data and the frequent emergence of new entities or events, effectively addressing the challenges of few-shot learning scenarios prevalent in the financial domain.
- We conduct extensive evaluations of our framework on real-world financial datasets, covering multiple tasks, and further validate cross-market transfer on an additional NASDAQ dataset. Experimental results demonstrate the superior performance, adaptability, and generalization capabilities of FGMPL.

The rest of our article is organized as follows: Section 2 reviews the related techniques and the state of corporate profiling. Section 3 introduces the dataset and problem definition. Section 4 details our proposed FGMPL method. Section 5 presents multi-task experiments on real-world financial data, demonstrating the effectiveness of our approach. Finally, we conclude the article in Section 6.

2 Related Work

2.1 Graph Pre-training

Graph pre-training technology, as a powerful paradigm, utilizes self-supervised learning on unlabeled graphs to capture intrinsic graph properties. Various methods design self-supervised tasks based on different graph attributes, including node features [35, 36], links [33, 90], and local or global patterns [53, 60]. For heterogeneous graphs, due to their complex node and edge types, specialized pre-training strategies [36, 51, 81] are required to effectively capture these intricate relationships. Strategies can be broadly categorized into generative and contrastive methods, with contrastive approaches typically outperforming generative ones. For example, GraphCL [91] pre-trains by minimizing the representational distance between the same graph under different augmentations. Consequently, contrastive methods are adopted as the default strategy in this article. However, pre-training aims to extract intrinsic knowledge from graphs in a self-supervised manner, while downstream tasks focus on specific supervised signals. This objective gap may hinder effective knowledge transfer, thereby impacting downstream performance.

2.2 Prompt-based Learning on Graphs

Graph prompt learning has emerged as a promising approach to bridge the gap between pre-training and downstream tasks in graph-based machine learning [30, 50, 77]. Prompts can be tailored for each downstream task, guiding each task towards the pre-trained model while keeping pre-trained parameters unchanged [78, 79]. Recent studies have explored graph prompt learning from different perspectives. GPPT [66] adopts link prediction as a pre-training task and designs prompts specifically for node classification. Both GraphPrompt [92] and ProG [68] aim to bridge the gap between pre-training and multiple downstream tasks, but GraphPrompt unifies tasks via a subgraph-similarity template with READOUT-level prompting, whereas ProG unifies tasks by reformulating node/edge tasks into induced-subgraph graph-level instances to better match graph-level pre-training. GraphPrompt+ [92] further generalizes GraphPrompt by adopting layer-wise prompts, i.e., injecting prompts across different encoder layers to exploit hierarchical representations beyond a single READOUT-stage prompt. EdgePrompt [30] introduces edge-level prompts by learning prompt vectors on edges and aggregating them during message passing. Furthermore, HGPrompt [54] extends GraphPrompt to heterogeneous graph learning, DyGPrompt [93] to dynamic graph learning, and ProNoG [94] to non-homophilic graph pre-training and prompt learning. Currently, graph prompt learning is primarily applied to node or graph classification tasks in open source benchmarks, with limited practical applications. The potential of graph prompt learning in various real-world applications remains largely unexplored.

2.3 Multi-task in Corporate Profiling

Corporate performance prediction remains challenging due to its volatility and non-stationarity [3]. Early research primarily employed technical analysis, utilizing historical price signals for trend prediction [21, 40, 73, 97, 99]. However, relying solely on price signals fails to capture the impact of market surprises. The Efficient Market Hypothesis [25] suggests that financial markets are informationally efficient, with stock prices reflecting all available information. Based on this theory, studies have expanded to examine the influence of news and social media on stock markets [6, 14, 46]. Yet, these non-graph approaches overlook the interdependencies among stocks. To address this limitation, researchers have adopted graph structures to represent stock relationships, transforming multi-stock trend prediction into a node classification problem solvable by GNN models [16, 98]. Recent studies have further advanced by integrating historical prices, social media

information, and stock relationships [37, 41, 47, 61] to achieve more comprehensive and accurate predictions.

Understanding corporate relationships is crucial for strategic decisions in supply chain management and partner selection [31], while financial literature highlights that stock risks and returns propagate through trading and supply chain networks [1]. Existing studies extract news and construct corporate network relationships, investigating their applicability in investment decision-making and corporate performance prediction [15, 55, 84]. After constructing corporate relationships as a graph structure, predicting corporate relationships can be transformed into a link prediction problem. Recent advances in GNNs have shown promising results in this domain, such as RGCN's attention mechanisms for modeling cooperative and competitive relationships [19], and SPGCN's decomposition of complementary and substitutive relationships [18], demonstrating the power of deep learning approaches in capturing complex corporate interactions.

In the information age, real-time news is widely published online, making it increasingly challenging for investors and corporate decision-makers to process daily news and identify potential connections. Traditional news corpus visualization approaches, without pre-specified queries, organize content through event timelines [74, 76, 85], event threads [59], event evolution graphs [86], or information maps [62, 63, 75, 83]. Recent studies leverage external knowledge (such as entity or story labels) to train embedding and clustering models [49, 57, 71]. Meanwhile, unsupervised methods [4, 45, 89] have adopted more practical approaches to adapt to evolving news content. Notably, these tasks primarily rely on textual content, with some research focusing on user propagation networks [8, 13, 72]. However, in-depth analysis of relationships between news subjects remains insufficient. In fact, news about different entities may reveal significant connections, warranting particular attention in news chain construction.

Different from existing works that handle single tasks with homogeneous information, we propose a unified multi-task framework FGMPL that leverages graph prompt learning and mixture-of-experts to effectively process heterogeneous financial information and adapt to various tasks through meta-learning enhanced prompt tuning.

3 Preliminaries

In this section, we introduce the real-world datasets used in our study, describe the pre-processing steps, explain the construction of the heterogeneous graph, and formally define the tasks addressed in this work.

3.1 Data Description

News Data. News articles were collected from five prominent Chinese financial news sources: China Securities Journal,¹ Shanghai Securities News,² Yicai,³ 21st Century Business Herald,⁴ and Securities Daily.⁵ The data collection period spanned from September 30, 2023, to December 31, 2023. Each news item includes its original text and publication timestamp. Figure 2(a) illustrates the daily news counts, revealing a periodic pattern aligned with market activity (e.g., opening and closing days).

Corporate Data. Corporate information was sourced from WIND,⁶ a leading financial information service provider in China. We extracted data for all Chinese listed companies from the WIND

¹<https://www.cs.com.cn/>.

²<https://www.cnstock.com/>.

³<https://www.yicai.com/>.

⁴<https://www.21jingji.com/>.

⁵<http://www.zqrb.cn/>.

⁶<https://www.wind.com.cn/>.

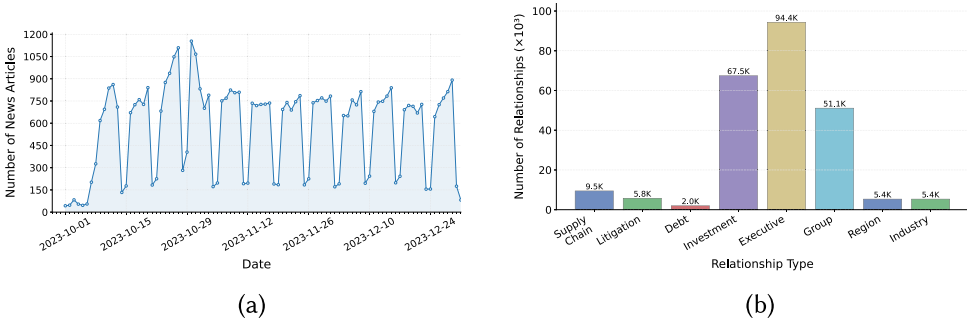


Fig. 2. (a) Daily news statistics. (b) Distribution of corporate relationship types.

database. The corporate dataset encompasses a wide array of relational information, including supply chain relationships, litigation associations, debt connections, shareholder investment links, and executive and group affiliations (see Figure 2(b) for the distribution of relationship types). While our primary focus is on 5,356 listed companies, our corporate graph also incorporates other firms to construct a comprehensive relational network and facilitate information propagation. Additionally, we also collect fundamental information such as industry classification, geographic location, and stock price.

3.2 Data Pre-processing

News Data Pre-processing. To address the issue of duplicate news articles reporting the same event, we implemented a TF-IDF-based cosine similarity approach. The similarity threshold was set to 0.8 based on empirical testing. When duplicate articles were identified, we retained the earliest published version to maintain temporal accuracy while avoiding information redundancy. For entity extraction from the news articles, we utilized Qwen-7B [9], a large language model specifically tailored for Chinese text processing and widely recognized in the field. This process identified companies, industries, and regions mentioned within the news content. In addition, the news articles were categorized into 20 classes using existing classification combined with a manual label by multiple experts. In our approach to encoding news events, we employ a dual strategy to capture both textual and temporal information. For the textual content, we utilize BERT [23] to generate rich, contextual representations of the news text (BERT uses a pre-trained model bert-base-Chinese⁷). For the temporal aspect, we convert each timestamp to an OLE date format [13], whose fractional and integral components form a 2D vector.

Corporate Data Pre-processing. In investment decision-making within financial markets, predicting price movement rather than directly forecasting stock prices is of critical importance. Price movement can more intuitively reflect the overall market trend and direction, thereby assisting investors in capturing the long-term development of the market rather than merely focusing on short-term price fluctuations [6, 41, 101]. Therefore, in this article, to evaluate company performance, we follow the rule of stock price fluctuations by calculating the stock performance relative to the market benchmark (CSI 300 Index) within 3 months from the reporting date [24, 27, 37, 88]. The performance indicator is computed as follows:

$$\text{Perf} = \frac{P_{t+3m} - P_t}{P_t} - \frac{I_{t+3m} - I_t}{I_t}, \quad (1)$$

where:

⁷<https://huggingface.co/google-bert/bert-base-chinese>.

Table 1. Statistics of Different Views

Data	Node	Node type	Edge	Edge type
News graph	44,230	4	96,676	1
Corporate graph	121,094	3	230,132	8
All graph	161,298	4	341,013	9

- P_t : Company stock price at report date.
- P_{t+3m} : Company stock price after 3 months.
- I_t : CSI 300 Index at report date.
- I_{t+3m} : CSI 300 Index after 3 months.

The classification rules can be formally expressed as:

$$\text{Classification} = \begin{cases} \text{Up,} & \text{if } Perf > +5\%, \\ \text{Neutral,} & \text{if } -5\% \leq Perf \leq +5\%, \\ \text{Down,} & \text{if } Perf < -5\%. \end{cases} \quad (2)$$

With regard to corporate relationship data, information on executives, groups, regions, and industries remains relatively stable in the short-term, which allows it to play a supplementary role in relationship prediction tasks by enriching network structures and facilitating effective information flow. In light of this, our study focuses on forecasting supply chain relationships, litigation, debt, and investment, as these factors significantly influence decision-making processes.

3.3 Heterogeneous Graph Construction

The integration of news and corporate data results in a comprehensive news-corporate heterogeneous graph that captures the complex relationships between entities. Based on the extracted entities, we constructed a news graph representing the interconnections between different news items and their associated entities. So the news graph consists of four node types—news articles, companies, industries, and regions—and one edge type representing “mention” relationships between news articles and entities. This graph captures the contextual and relational information inherent in financial news. Utilizing the relational data, we constructed a corporate relationship graph. The graph consists of three types of nodes: entities, regions, and industries, with eight types of edges, including the aforementioned relationships as well as regional and industrial affiliations. Finally, to create a unified representation, we merged the news graph and corporate graph by linking entities (e.g., companies, industries, and regions) appearing in both datasets. The result is a comprehensive news-corporate heterogeneous graph, which serves as the foundation for our subsequent analytical tasks. The statistics of each graph are shown in Table 1.

A news-corporate heterogeneous graph can be formally defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where: $\mathcal{V} = \mathcal{V}_N \cup \mathcal{V}_C$ is the set of nodes, with $\mathcal{V}_N$ representing the set of news nodes and $\mathcal{V}_C$ representing the set of corporate nodes. $\mathcal{E} = \mathcal{E}_{CC} \cup \mathcal{E}_{NC}$ is the set of edges, representing corporate-corporate edge set and news-corporate edge set, respectively. The node type mapping function is defined as $\phi: \mathcal{V} \rightarrow \mathcal{A}$, where $\mathcal{A} = \{a_N, a_C\}$, with a_N denoting the news node type and a_C denoting the corporate node type. The edge type mapping function is defined as $\psi: \mathcal{E} \rightarrow \mathcal{R}$, where $\mathcal{R} = \{r_{CC}, r_{NC}\}$, representing corporate-corporate edge type and news-corporate edge type, respectively. This definition satisfies the heterogeneous graph condition of $|\mathcal{A}| + |\mathcal{R}| > 2$.

The use of a heterogeneous graph structure provides several advantages for modeling the news-corporate network. Its ability to encode diverse entities and relationships makes it well-suited

for capturing the complexity inherent in this data. By utilizing the message-passing mechanisms of GNNs, it allows efficient propagation and integration of multi-source information, such as news semantics, corporate attributes, and relational structures. The structural characteristics of heterogeneous graphs further enable the discovery of indirect relationships and higher-order connections, which are critical for uncovering latent corporate relationships and predicting corporate performance. Moreover, the explicit differentiation of node and edge types offers strong interpretability, allowing insights into the relative importance of various factors in predictive tasks. Collectively, these properties make the heterogeneous graph an effective and versatile framework, capable of supporting multi-task learning by enabling knowledge sharing and accommodating diverse downstream tasks at both node and edge levels.

Besides the Chinese market datasets, we also construct an additional NASDAQ heterogeneous news–corporate graph for the cross-market transfer study (reported in Section 5.3). We follow the same heterogeneous graph schema introduced in this section and merge the news graph and the corporate graph by aligning shared entities. Basic statistics of the NASDAQ dataset are summarized in Table 5.

3.4 Task Definition

Given a news–corporate heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \mathcal{V}_N \cup \mathcal{V}_C$ represents news and corporate nodes, and $\mathcal{E} = \mathcal{E}_{CC} \cup \mathcal{E}_{NC}$ represents corporate–corporate and news–corporate edges, respectively, our goal is to solve three interconnected tasks:

- (1) *Corporate Performance Prediction*: For any corporate node $v_c \in \mathcal{V}_C$, predict its performance category $y_c \in \{Up, Neutral, Down\}$ based on its relative market performance defined in Equation (1).
- (2) *Corporate Relationship Prediction*: For any corporate–corporate edge $e_{cc} \in \mathcal{E}_{CC}$ between two corporate nodes, predict its relationship type $y_r \in \{None, Supply, Debt, Litigation, Investment\}$.
- (3) *News Classification*: For any news node $v_n \in \mathcal{V}_N$, classify its content into 1 of 20 predefined categories $y_n \in \{m_1, m_2, \dots, m_{20}\}$.

4 Proposed Approach: FGMPL

In this section, we present FGMPL, an innovative approach designed to better understand and analyze the complex interactions within financial ecosystems (shown in Figure 3). Our method addresses the challenges through four key components: a unified task format, multi-view pre-training, Financial Multi-Experts Prompting mechanism, and meta-learning enhancement. First, we introduce a unified graph-level instance representation method that integrates various corporate profiling tasks into a coherent framework. Notably, this unified formulation is implemented via local induced subgraphs centered at each target node or edge, rather than collapsing the entire financial graph into a single global representation, thereby preserving fine-grained neighborhood structure and task-specific interactions. Second, we employ multi-view contrastive learning for pre-training, which effectively captures both explicit relationships (such as supply chain connections and ownership structures) and implicit relationships (derived from news co-mentions and market behavior correlations) in the financial ecosystem. Third, we propose an innovative Financial Multi-Experts Prompting mechanism, which uniquely integrates prompt templates with a MoE architecture, enabling specialized experts to collaboratively process distinct aspects of heterogeneous financial information and effectively address the complexities of financial data. Finally, we integrate meta-learning techniques into our prompt tuning process, enabling rapid adaptation to

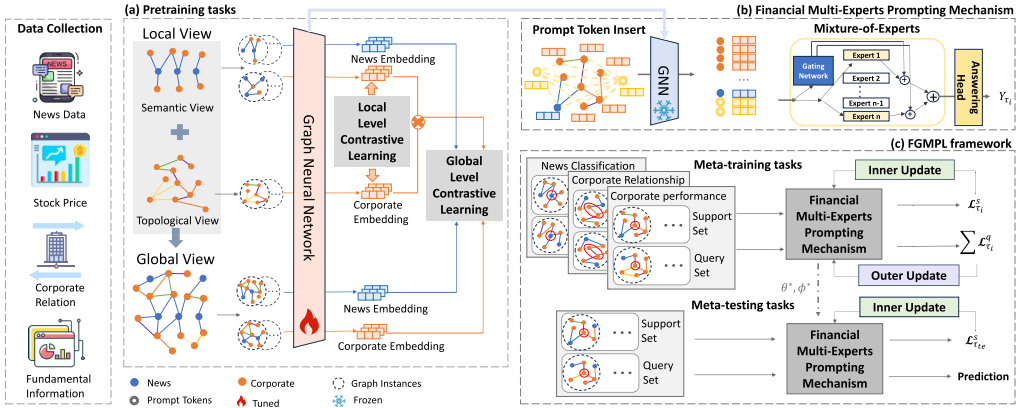


Fig. 3. An overview of our FGMPL.

new financial scenarios and tasks. In the following subsections, we will elaborate on the specific implementation and theoretical foundations of each component.

4.1 Unified Task Format

Unified Graph-level Instances. Most previous works have not attempted to solve both node and edge tasks of a corporate at the same time, mainly because in the graph domain, node-level, edge-level, and graph-level tasks exhibit significant differences. This gap limits the performance of pre-trained models and may even lead to negative transfer. However, fundamental graph operations—such as modifying node features or adding and removing edges—can be interpreted as elementary operations at the graph level. Therefore, following [68], we reformulate both node-level and edge-level tasks as graph-level tasks by constructing induced graphs for nodes and edges, respectively. We do not aggregate the entire financial graph into one coarse representation; instead, for each target node/edge, we construct a local induced subgraph and treat it as a graph-level instance, which preserves the fine-grained neighborhood structure required by node-/edge-centric tasks. Moreover, the Financial Multi-Experts mechanism introduced in Section 4.3 further reduces the risk that localized dependencies and other fine-grained details are diluted during aggregation, by allowing different experts to focus on different structural or semantic aspects of the same induced subgraph instead of enforcing a single shared representation.

At the node level, such as a corporate v , we typically define the local neighborhood of the corporate v in graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ as a context subgraph $S_v = (\mathcal{V}(S_v), \mathcal{E}(S_v))$, where the node set and edge set are given by:

$$\mathcal{V}(S_v) = \{u \in \mathcal{V} \mid d(u, v) \leq \delta\}, \quad (3)$$

$$\mathcal{E}(S_v) = \{(u, u') \in \mathcal{E} \mid u \in \mathcal{V}(S_v), u' \in \mathcal{V}(S_v)\}. \quad (4)$$

Here, $d(u, v)$ denotes the shortest distance between corporates u and v in graph $\mathcal{G}$, and δ is a predetermined threshold. In other words, S_v consists of corporates within δ hops of corporate v and the relationships between these corporates. This subgraph preserves the local structure of the node through adjacent node connections and its semantic context through adjacent node features. It means the corporate v with some most related corporate and news. By assigning the target node's label to its corresponding induced graph, we can readily recast the node classification problem as a graph classification task. Similarly, at the edge level, we obtain an induced graph for a pair of corporates. This subgraph is constructed by extending the pair of corporates to include their

neighbors within a distance of τ . Edge-level tasks can thus be reformulated by assigning the induced graph's label as the edge label corresponding to the target node pair.

Unified Task Template. Building upon the aforementioned subgraph definitions for node- and edge-level instances, we can unify various tasks into a common template. Let $\mathbf{s}_x$ denote the vector representation of subgraph S_x , and $\text{sim}(\cdot, \cdot)$ represent the cosine similarity function. These tasks can be mapped to the computation of subgraph similarity, which we formalize as follows:

- Node Classification: *Corporate Performance Prediction* and *News Classification* can be seen as node classification task. Consider a news–corporate heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with a set of node classes C (which is $\{Up, Neutral, Down\}$ for corporate nodes and $\{m_1, m_2, \dots, m_{20}\}$ for news nodes), and a set of labeled nodes $D = \{(v_1, \ell_1), (v_2, \ell_2), \dots\}$, where $v_i \in V$ and ℓ_i is the corresponding label of v_i . In a k -shot setting, there are exactly k pairs of $(v_i, \ell_i = c) \in D$ for each class $c \in C$. For each class $c \in C$, we define a node class prototypical subgraph, represented by vector $\tilde{\mathbf{s}}_c$:

$$\tilde{\mathbf{s}}_c = \frac{1}{k} \sum_{(v_i, \ell_i) \in D, \ell_i = c} \mathbf{s}_{v_i}. \quad (5)$$

Note that the class prototypical subgraph is a “virtual” subgraph with a latent representation in the same embedding space as the contextual subgraphs. It is constructed as the mean representation of the contextual subgraphs of labeled nodes in a given class. Then, for a node v_j not in the labeled set D , its class label ℓ_j should be:

$$\ell_j = \arg \max_{c \in C} \text{sim}(\mathbf{s}_{v_j}, \tilde{\mathbf{s}}_c). \quad (6)$$

- Link Prediction: *Corporate Relationship Prediction* can be seen as a link prediction task. Given a news–corporate heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with corporate relationship types $R = \{None, Supply, Debt, Litigation, Investment\}$, and a set of labeled corporate relationships $D_e = \{((u_1, v_1), r_1), ((u_2, v_2), r_2), \dots\}$ where $(u_i, v_i) \in \mathcal{E}$ and $r_i \in R$ denotes the corresponding edge type. In a k -shot setting, there are exactly k pairs of $((u_i, v_i), r_i = r) \in D_e$ for each relationship type $r \in R$. For each relationship type $r \in R$, we define a relationship type prototypical subgraph, represented by vector $\tilde{\mathbf{s}}_r$:

$$\tilde{\mathbf{s}}_r = \frac{1}{k} \sum_{((u_i, v_i), r_i) \in D_e, r_i = r} \mathbf{s}_{(u_i, v_i)}. \quad (7)$$

Then, for a candidate corporate relationship (u_j, v_j) not in the labeled set D_e , its relationship type r_j should be:

$$r_j = \arg \max_{r \in R} \text{sim}(\mathbf{s}_{(u_j, v_j)}, \tilde{\mathbf{s}}_r). \quad (8)$$

To obtain the subgraph embedding vector $\mathbf{s}_x$ for subgraph S_x , we utilize node representations $\mathbf{h}_v$ computed by a GNN (as described in Section 4.2). The standard approach involves applying a READOUT operation to aggregate node representations within S_x :

$$\mathbf{s}_x = \text{READOUT}(\{\mathbf{h}_v : v \in \mathcal{V}(S_x)\}). \quad (9)$$

For the READOUT aggregation function, various options exist, including sum pooling and more sophisticated methods [82, 87]. During the pre-training phase, we employ sum pooling, with further discussion provided in Section 5.4.

4.2 Multi-view Pre-training

To effectively capture the complex interactions in our heterogeneous news–corporate network, we employ RGAT [12], which specifically considers different types of relationships while computing node representations through message passing. Given a corporate or news v_i with initial features $\mathbf{h}_i$, we update its representation by aggregating messages from its neighbors while considering the relationship types. The message passing process can be formulated as:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \alpha_{ij}^r \mathbf{W}_r^{(l)} \mathbf{h}_j^{(l)} \right), \quad (10)$$

where $\mathbf{h}_i^{(l)}$ represents the node features at layer l , $\mathcal{R}$ is the set of relation types, $\mathcal{N}_i^r$ denotes the neighbors of corporate i connected by relationship type r , $\mathbf{W}_r^{(l)}$ is a relationship-specific transformation matrix, and α_{ij}^r represents the attention coefficient for edge (i, j) with relationship type r . The attention coefficients α_{ij}^r are computed using multi-head attention to stabilize the learning process:

$$\alpha_{ij}^r = \text{softmax}_j(e_{ij}^r) = \frac{\exp(e_{ij}^r)}{\sum_{k \in \mathcal{N}_i^r} \exp(e_{ik}^r)}, \quad (11)$$

where e_{ij}^r is calculated as:

$$e_{ij}^r = \text{LeakyReLU}(\mathbf{a}_r^T [\mathbf{W}_r \mathbf{h}_i | \mathbf{W}_r \mathbf{h}_j]). \quad (12)$$

Here, $\mathbf{a}_r$ is a learnable attention vector specific to relation type r , and $|$ denotes concatenation.

In pre-training, we employ multi-view contrastive learning to leverage the heterogeneity and complexity of our news–corporate data and obtain more comprehensive and enriched node representations. Specifically, news nodes primarily focus on textual content, which contains rich semantic information, while corporate nodes emphasize their complex relationships and structural positions within the commercial network. Additionally, the global-level perspectives, news–corporate graph provides cross-domain integrated information, including direct connections and potential long-range dependencies between news and corporate. Through contrastive learning across these different views, the model can learn node representations that incorporate both local details and global structures, thereby providing more effective features for downstream tasks.

The contrastive learning process leverages these multiple views to learn enriched node embeddings. For each view, given a corporate or news i , we compute its subgraph representations in three views: (1) Semantic view: $\mathbf{s}_i^{\text{sem}}$ through textual content. (2) Topological view: $\mathbf{s}_i^{\text{topo}}$ through structural information. (3) Global view: $\mathbf{s}_i^{\text{all}}$ through the complete graph. Each representation is obtained by applying the READOUT function to the corresponding subgraph:

$$\mathbf{s}_i^{\text{view}} = \text{READOUT}(\mathbf{h}_v^{\text{view}} : v \in \mathcal{V}(S_i)), \quad (13)$$

Local Level Contrastive Learning. We first focus on the local level, where corporate nodes exist in both semantic and topological views. Contrastive learning aligns these views, enhancing node representations to incorporate context-driven and structure-driven features, thereby improving generalization and capturing multi-view shared characteristics.

Specifically, for each batch of corporate nodes, we generate embeddings $\mathbf{s}_i^{\text{sem}}$ and $\mathbf{s}_i^{\text{topo}}$ from their respective GNNs. The local contrastive loss $\mathcal{L}_{\text{local}}$ is defined as:

$$\mathcal{L}_{\text{local}} = -\log \frac{\exp(\text{sim}(\mathbf{s}_i^{\text{sem}}, \mathbf{s}_i^{\text{topo}})/\tau)}{\sum_j \exp(\text{sim}(\mathbf{s}_i^{\text{sem}}, \mathbf{s}_j^{\text{topo}})/\tau)}, \quad (14)$$

where $\text{sim}(\cdot, \cdot)$ denotes the similarity function (e.g., cosine similarity), and τ is the temperature parameter.

Global Level Contrastive Learning. Following the local contrastive learning, we further design a global contrastive learning objective to integrate information at a higher level. We first fuse the semantic and topological embeddings into a unified representation $\hat{s}_i = f(s_i^{\text{sem}}, s_i^{\text{topo}})$, where $f(\cdot, \cdot)$ is a fusion function (e.g., concatenation, summation, or more complex methods; we use summation). This fused embedding $\hat{s}_i$ combines the strengths of both semantic and topological information. We then contrast this fused embedding with the global-level embedding s_i^{all} , which is generated from the entire graph through message passing. The global-level embedding integrates cross-view information, aligning local representations with global constraints to enhance their consistency and robustness. The global contrastive loss for corporate $\mathcal{L}_{\text{global}}$ is defined as:

$$\mathcal{L}_{\text{global}} = -\log \frac{\exp(\text{sim}(\hat{s}_i, s_i^{\text{all}})/\tau)}{\sum_j \exp(\text{sim}(\hat{s}_i, s_j^{\text{all}})/\tau)}. \quad (15)$$

At the same time, we introduce contrastive learning between news nodes and the global view, aiming to strengthen the relationship between news information and the overall graph. This enhances news representations' relevance to enterprises, improving the model's sensitivity to news-enterprise interactions and downstream task performance.

The contrastive loss for news and global node representations is formulated as:

$$\mathcal{L}_{\text{news}} = -\log \frac{\exp(\text{sim}(s_i^{\text{news}}, s_i^{\text{all}})/\tau)}{\sum_j \exp(\text{sim}(s_i^{\text{news}}, s_j^{\text{all}})/\tau)}. \quad (16)$$

The overall pre-training loss is a combination of the corporate contrastive losses and the news contrastive loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{local}} + \lambda_1 \mathcal{L}_{\text{global}} + \lambda_2 \mathcal{L}_{\text{news}}, \quad (17)$$

where λ_1 and λ_2 are coefficients for the respective losses. By minimizing this loss, the model learns representations that effectively combine local and global information from different perspectives, improving the quality of embeddings for downstream tasks. This multi-view pre-training is performed offline to learn a general-purpose encoder. During downstream training, we freeze the pre-trained GNN encoder and update only the prompt tokens, MoE parameters, and task-specific heads, which constrains the optimization space, stabilizes training, and reduces the risk of prompt tokens drifting toward degenerate solutions. At inference time, we reuse the same frozen encoder, so the multi-view pre-training does not introduce additional inference-time complexity beyond a standard forward pass.

4.3 Financial Multi-experts Prompting Mechanism

4.3.1 Prompt Design. For graph prompting applications in the financial domain, we propose an innovative prompt token design scheme. Through in-depth analysis of recent advances in graph representation learning, we discover that graph prompting can effectively bridge news-corporate multi-view pre-training with diverse downstream financial tasks. Previous works have proposed different approaches to graph prompting design. The most fundamental method involves adding additional features to the original graph feature matrix $\mathbf{X} = \{x_1, \dots, x_N\} \in \mathbb{R}^{N \times d}$. For instance, [26, 100] employ a learnable vector $\mathbf{p} \in \mathbb{R}^{1 \times d}$ as the basic prompt, superimposing it onto all node features to form a new feature matrix $\mathbf{X}^* = \{x_1 + \mathbf{p}, \dots, x_N + \mathbf{p}\}$ (Figure 4(a)). GraphPrompt [92] transfers prompt token design to the hidden layers of the graph model, achieving a more compact representation through dimensionality reduction and employing element-wise multiplication for

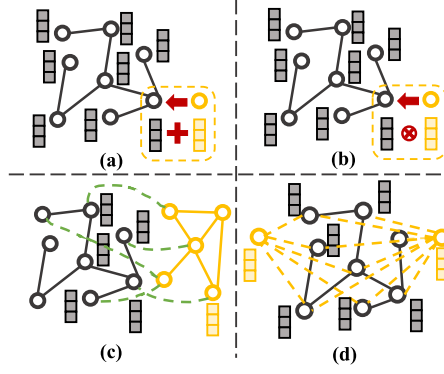


Fig. 4. Comparison of graph prompting designs. (a) By feature adding; (b) By multiplication in hidden layers; (c) Prompt as graphs with cross-links; (d) Multiple globally-connected prompt tokens.

feature fusion (Figure 4(b)). Furthermore, GraphPrompt+ extends GraphPrompt beyond readout-/single-layer prompting by introducing layer-wise prompts, i.e., a set of learnable prompt vectors injected in parallel into the input layer and each hidden layer, to explicitly leverage the hierarchical representations across GNN layers. ProG [68] advances further by designing graph prompts as complete subgraph structures, facilitating information exchange with the original graph through internal and cross-linking connections (Figure 4(c)).

Our approach constructs multiple globally connected prompt tokens (Figure 4(d)). This design maintains a holistic perception of the news-corporate graph while effectively integrating specific financial task information. Compared to methods that simply add extra features, our design offers enhanced flexibility and expressiveness. Relative to subgraph prompting, our approach significantly reduces introduced edge relationship noise while lowering computational complexity. Specifically, we introduce a set of learnable prompt tokens $\mathcal{T} = \tau_1, \tau_2, \dots, \tau_{|\mathcal{T}|}$, where each token τ_i is represented by a vector $\mathbf{t}_i \in \mathbb{R}^{1 \times d}$, with d being the dimension of node features in the input graph. These tokens are added to the original graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, forming an augmented graph $\mathcal{G}_\tau = (\mathcal{V} \cup \mathcal{T}, \mathcal{E} \cup \mathcal{Q})$, where $\mathcal{Q}$ represents newly formed edges between prompt tokens and original nodes. The prompt tokens participate in the message passing process of the GNN, enabling them to capture and contribute task-specific information. For a given node v_i with original features $\mathbf{x}_i$, its representation is updated through message passing with connected prompt tokens during the GNN computation:

$$\mathbf{H}^{(l+1)} = \text{GNN}(\mathbf{X}_{\text{aug}}, \mathcal{G}_\tau), \quad (18)$$

where $\mathbf{X}_{\text{aug}} \in \mathbb{R}^{(N+|\mathcal{T}|) \times d}$ is the augmented feature matrix containing both original node features and prompt token features, and $\mathbf{H}^{(l+1)}$ represents the node representations at layer $l + 1$ of the GNN. All prompt tokens are initialized from a zero-mean Gaussian distribution with small variance, following common practice in prompt-based learning, to avoid introducing strong prior bias at the early training stage.

4.3.2 Financial Multi-experts. After incorporating prompt tokens, we propose Financial Multi-Experts Prompting mechanism to enhance the prompt by introducing specialized expert components that reflect the diverse roles and interactions within financial ecosystems. Similar to how different market participants (analysts, traders, regulators) bring unique perspectives to financial markets, each expert network specializes in processing specific aspects of market information and corporate relationships. First, we implement specialized expert networks, analogous to different types of financial analysts or market participants. Each expert network $f_i(\mathbf{x})$ processes information through

a two-layer structure:

$$f_i(\mathbf{x}) = W_2^i(\text{ReLU}(W_1^i \mathbf{x} + b_1^i)) + b_2^i, \quad (19)$$

where $W_1^i \in \mathbb{R}^{d \times d}$, $W_2^i \in \mathbb{R}^{1 \times d}$ are learnable parameters, b_1^i and b_2^i are bias terms, and d is the input dimension.

The specialized networks function as analysts focusing on distinct analytical domains, such as fundamental analysis, technical analysis, or news sentiment analysis. Subsequently, mirroring how market participants selectively attend to the most relevant information sources. We follow the strategy that achieves sparsity by computing a weighted sum of the outputs from only the top- k experts, rather than aggregating the outputs from all the experts [65]. So the gating network can be formulated as:

$$\text{Gate}(\mathbf{x}; \Theta)_i = \text{softmax}(\text{TopK}(g(\mathbf{x}; \Theta), k))_i, \quad (20)$$

$$\text{TopK}(g(\mathbf{x}; \Theta), k)_i = \begin{cases} g(\mathbf{x}; \Theta)_i, & \text{condition,} \\ -\infty, & \text{otherwise,} \end{cases} \quad (21)$$

$$\text{condition : if } g(\mathbf{x}; \Theta)_i \text{ is in the top-}k \text{ elements of } g(\mathbf{x}; \Theta). \quad (22)$$

where Θ represents the learnable parameters of the gating network, and k is the number of selected experts.

The selection and weighting process emulates the financial market's integration of diverse expert opinions, where certain perspectives may carry greater weight depending on specific market conditions. The final importance weights for tokens and original nodes are derived through the combination of expert opinions:

$$w_v = \sum_{i \in \text{TopK}} \text{Gate}(\mathbf{x}_v; \Theta)_i f_i(\mathbf{x}_v). \quad (23)$$

The graph-level representation is computed through weighted aggregation:

$$\mathbf{s}_G = \frac{\sum_{v \in \mathcal{V}} w_v \mathbf{x}_v}{\sum_{v \in \mathcal{V}} w_v}, \quad (24)$$

where $\mathcal{V}$ denotes the set of all nodes in the graph, including both original nodes and prompt tokens. Analogous to financial markets not requiring consultation with every analyst for each decision, our system activates only the most relevant experts for each input. We further visualize the resulting expert routing patterns and show that the learned routing is sparse and task-dependent (e.g., a task-level heatmap), providing an intuitive interpretation of how different experts contribute to different tasks (Section 5.6).

4.4 Prompt Tuning via Meta-learning

We propose a method to extend prompt tuning by integrating it with a meta-learning framework, which facilitates the optimization of prompt tokens and model parameter initialization for rapid adaptation across various financial scenarios. To effectively address the computational demands inherent in processing high-frequency financial data and real-time requirements, we implement a first-order approximation of the Model-Agnostic Meta-Learning algorithm [28]. Specifically, our meta-learning framework employs a two-level optimization process for prompt initialization. The outer optimization focuses on learning transferable base prompt tokens and model parameters to enable rapid task adaptation, while the inner optimization performs task-specific fine-tuning. Unlike [68], we include not only the answering head but also MoE-related parameters in task-specific optimization. This enhances the model's adaptability to multi-task and fine-grained feature distributions, thereby improving its generalization performance and task robustness.

For each task τ_i , we define a support set $D_{\tau_i}^s$ and a query set $D_{\tau_i}^q$. Let θ be the prompt parameters, π^* be the fixed parameters of the pre-trained graph backbone, and ϕ be the task-specific parameters. The model pipeline is denoted as $f_{\theta, \phi | \pi^*}$. For task τ_i , the loss function is defined as:

$$\mathcal{L}_{\tau_i}(D) = -\frac{1}{|D|} \sum (x, y) \in D y \log(f_{\theta, \phi | \pi^*}(x)), \quad (25)$$

where y represents the ground-truth label (e.g., actual corporate relationship type), and $f_{\theta, \phi | \pi^*}(x)$ denotes the predicted class probabilities output by the model. During the inner loop optimization, for each task τ_i , we update the parameters using one gradient step:

$$\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\tau_i}(D_{\tau_i}^s), \quad (26)$$

$$\phi'_i = \phi - \alpha \nabla_{\phi} \mathcal{L}_{\tau_i}(D_{\tau_i}^s), \quad (27)$$

where α is the inner loop learning rate. The meta-objective function is defined as finding optimal initialization settings (θ, ϕ) for the prompt tokens and task-specific parameters:

$$\theta^*, \phi^* = \arg \min_{\theta, \phi} \sum_{\tau_i \in \mathcal{T}} \mathcal{L}_{\tau_i}(D_{\tau_i}^q). \quad (28)$$

To optimize the meta-objective, we perform outer loop updates using stochastic gradient descent across all tasks:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_{\tau_i \in \mathcal{T}} \mathcal{L}_{\tau_i}(D_{\tau_i}^q; f_{\theta'_i, \phi'_i | \pi^*}), \quad (29)$$

$$\phi \leftarrow \phi - \beta \nabla_{\phi} \sum_{\tau_i \in \mathcal{T}} \mathcal{L}_{\tau_i}(D_{\tau_i}^q; f_{\theta'_i, \phi'_i | \pi^*}), \quad (30)$$

where β is the outer loop learning rate. The simultaneous optimization of inner loop and outer loop objectives enables the model to learn an effective initialization state for rapid adaptation to new tasks within the financial ecosystem. The meta-learning framework demonstrates particular advantages in handling the dynamic nature of financial markets, such as the introduction of new companies, emergence of novel corporate relationships, and rapid shifts in market sentiment, requiring only a few gradient steps for effective adaptation.

In our framework, the prompt tokens are trained with a two-level meta-learning objective: the outer loop learns transferable base prompt initialization shared across tasks/graphs, while the inner loop performs task-specific adaptation with a few gradient steps. We further constrain the optimization by keeping the pre-trained graph backbone fixed and only updating lightweight prompt-related parameters, which reduces the chance of drifting to degenerate prompt solutions. In addition, the sparse top-k MoE routing limits each instance to only a small subset of experts, so less helpful experts receive negligible gating weights and have limited influence, mitigating suboptimal prompt convergence under heterogeneous distributions or sparsely connected graphs. We further validate these designs empirically in section 5.

5 Experiments

In this section, we will introduce the details of our experiment on real-world datasets to validate the efficiency of our proposed FGMPL.

5.1 Experimental Settings

Datasets. For our experiments, we adopt different data splitting strategies for full-data and few-shot settings. The details regarding dataset collection and processing have been discussed in Sections 3.1 and 3.2. In the full-data setting, news classification follows a temporal split using articles from

2023-9-30 to 2023-11-30 for training (70%), 2023-12-1 to 2023-12-15 for validation (15%), and the remaining December data for testing (15%). For corporate-related tasks, we randomly split corporate nodes into training (70%), validation (15%), and test (15%) sets while maintaining graph connectivity and relationship distribution. In the few-shot scenario, following [68], we adopt a 1:1 split ratio for meta-training and meta-testing sets across all tasks to evaluate the model's generalization ability to unseen samples.

Baselines. We compare our proposed method against approaches, which can be categorized into three main classes:

- (1) *End-to-end GNNs:* These methods directly train GNN models on specific tasks without a pre-training phase. We select several representative models for comparison, including GCN [42], GAT [69], and RGAT [12]. They capitalize on the key operation of neighborhood aggregation to recursively aggregate messages from the neighbors, and work in an end-to-end manner.
- (2) *Graph pre-training models:* These approaches follow the “pre-train, fine-tune” paradigm. We choose several typical models for comparison, such as DGI [70], GraphCL [91], and SimGRACE [80]. They work in the “pre-train, fine-tune” paradigm. In particular, they pre-train the GNN models to preserve the intrinsic graph properties, and fine-tune the pre-trained weights on downstream tasks to fit task labels.
- (3) *Graph prompt learning models:* These methods utilize learnable prompt graphs to reconstruct the input graph and reformulate downstream tasks to fit pre-training strategies. We select GPPT [51], GraphPrompt/GraphPrompt+ [92], and ProG [68] as representatives of this class. Since the pre-training objectives of GPPT and GraphPrompt/GraphPrompt+ are inherently edge-level prediction tasks, we do not include them as baselines in our corporate relationship prediction setting.
- (4) *Single-task models:* To comprehensively validate our framework, we compare it with models specialized for individual tasks. Specifically, for the corporate performance task, we select HAN [37] (utilizing news data) and HATS [41] (leveraging relational data); for the corporate relationship task, we adopt **Node2Vec (N2V)** [32] and **Variational Graph Auto-encoder (VGAE)** [43], both designed for link prediction; and for the news classification task, we use EventX [49] and SCStory [89], originally developed for news event chain generation. Notably, we only employ their preliminary components tailored for classification.

Settings and Parameters. To evaluate the efficacy of our FGMPPL methodology across various downstream tasks, we employed Accuracy (ACC) and F1-score as performance metrics. Each experiment was conducted five times, and the mean values of these metrics were calculated. Throughout our experimentation, we utilized a two-layer GNN with a hidden dimension of 128. While ProG adhered to the original configuration with 10 tokens, our approach was set to 5 tokens. Additionally, we conducted supplementary experiments, varying the token count from 1 to 10, to analyze its impact on performance (detailed results are presented in Section 5.5). We implemented a top-k MoE mechanism with an activation ratio of 0.2, and the number of experts, E , was set to 20. The learning rate was typically configured at 0.0001. To select the optimal hyperparameters λ_1 and λ_2 for our loss function, we employed a grid search method by exploring candidate values in the set $\{0.01, 0.1, 1, 10\}$ and ultimately chose the combination that achieved the best performance on the validation set. For baseline models, we followed their original parameter settings: GCN, GAT, and RGAT were implemented with two layers. For pre-training models (DGI, GraphCL, and SimGRACE), we used their default configurations with the same backbone architecture. For prompt learning models, GPPT, GraphPrompt/GraphPrompt+, and ProG were implemented with their recommended settings, using 10 learnable tokens. All baseline models were optimized using

Table 2. Performance Evaluation (%) on 100-Shot

Training schemes		Corporate performance		Corporate relationship		News classification	
Methods		ACC	F1	ACC	F1	ACC	F1
Supervised	GCN	59.45 $\pm$ 16.48	59.79 $\pm$ 4.62	28.44 $\pm$ 9.58	31.41 $\pm$ 10.58	19.09 $\pm$ 1.99	23.39 $\pm$ 2.82
	GAT	62.16 $\pm$ 12.42	60.21 $\pm$ 9.15	31.36 $\pm$ 2.11	29.66 $\pm$ 2.46	18.71 $\pm$ 4.24	22.24 $\pm$ 4.80
	RGAT	63.93 $\pm$ 9.30	62.06 $\pm$ 5.83	33.31 $\pm$ 5.52	32.40 $\pm$ 5.75	21.69 $\pm$ 4.60	22.45 $\pm$ 6.27
Pre-train	DGI	68.71 $\pm$ 6.56	67.13 $\pm$ 4.69	33.87 $\pm$ 3.79	32.16 $\pm$ 5.36	24.74 $\pm$ 3.01	22.90 $\pm$ 3.21
	GraphCL	69.55 $\pm$ 11.31	67.28 $\pm$ 7.06	35.93 $\pm$ 5.24	34.22 $\pm$ 6.73	24.57 $\pm$ 4.14	23.11 $\pm$ 3.10
	SimGRACE	69.92 $\pm$ 8.27	68.36 $\pm$ 7.94	34.42 $\pm$ 4.18	33.87 $\pm$ 4.69	25.30 $\pm$ 5.02	23.42 $\pm$ 4.09
Prompt	GPPT	64.65 $\pm$ 14.77	62.72 $\pm$ 10.05	-	-	23.69 $\pm$ 4.24	22.43 $\pm$ 2.97
	GraphPrompt	68.23 $\pm$ 11.85	66.42 $\pm$ 7.21	-	-	28.47 $\pm$ 6.12	30.58 $\pm$ 6.83
	GraphPrompt+	70.86 $\pm$ 8.42	69.21 $\pm$ 4.35	-	-	30.62 $\pm$ 4.27	32.94 $\pm$ 4.52
	ProG	71.48 $\pm$ 10.33	70.09 $\pm$ 5.87	44.07 $\pm$ 9.27	42.02 $\pm$ 11.31	31.04 $\pm$ 5.38	33.76 $\pm$ 5.98
	FGMPL	79.05 $\pm$ 3.74	77.99 $\pm$ 1.81	50.46 $\pm$ 7.70	47.59 $\pm$ 7.08	39.79 $\pm$ 3.34	43.28 $\pm$ 3.27
IMP		7.57%	7.90%	6.39%	5.57%	8.75%	9.52%

The best results are highlighted in bold.

Adam optimizer with a learning rate of 0.0001. Our model was implemented using PyTorch and PyTorch Geometric libraries. All experiments were conducted on an NVIDIA 3090 GPU with 24GB of memory. All prompt embeddings are initialized with a zero-mean Gaussian distribution with small variance and have the same feature dimension d as node embeddings. During downstream training, we freeze the pre-trained graph encoder and only optimize the prompt parameters, the MoE parameters, and task-specific heads using Adam. The prompt parameters are updated solely through the downstream loss and the meta-learning objective in Equations (25)–(30).

5.2 Performance Evaluation

As discussed, we perform three kinds of tasks, namely, corporate performance prediction, corporate relationship prediction and news classification in 100-shot settings, and then we extended our experiments to encompass the full dataset.

Downstream Tasks Performance with Few Shot. In a few-shot setting, we compared our prompt-based approach with other mainstream training paradigms on corporate performance prediction, corporate relationship prediction, and news classification tasks. We conducted five repetitions of the evaluation and report the average results in Table 2. Firstly, our proposed FGMPL outperformed all baselines across the three tasks, surpassing the next best method (ProG) by 7.57%, 6.39%, and 8.75%, respectively, demonstrating FGMPL’s efficacy in transferring knowledge from pre-training to multiple downstream tasks for superior performance. Secondly, we observed from the results that most supervised methods struggled to achieve better performance compared to pre-training and prompting approaches. This can be attributed to the limited empirical annotations required by supervised frameworks in few-shot settings, leading to suboptimal performance. In contrast, pre-training methods incorporate more prior knowledge, rendering graph models less dependent on data labels. Among supervised methods, RGAT consistently outperformed GCN and GAT across all tasks, indicating its architecture’s enhanced effectiveness in capturing task-specific graph structures when rich corporate relational information is available.

Downstream Tasks Performance with All Data. Following an in-depth analysis of the few-shot scenario, we extended our experimental scope to encompass full dataset evaluation. The results of these experiments are presented in Table 3. In the full-sample context, our proposed FGMPL

Table 3. Performance Evaluation (%) on All Data

Training schemes		Corporate performance		Corporate relationship		News classification	
Methods		ACC	F1	ACC	F1	ACC	F1
Supervised	GCN	76.51 $\pm$ 4.13	79.01 $\pm$ 2.64	58.83 $\pm$ 5.25	59.26 $\pm$ 3.10	45.94 $\pm$ 2.78	47.01 $\pm$ 4.23
	GAT	77.53 $\pm$ 2.82	80.06 $\pm$ 2.41	62.16 $\pm$ 4.37	61.49 $\pm$ 2.84	47.80 $\pm$ 3.31	49.12 $\pm$ 3.70
	RGAT	79.30 $\pm$ 4.29	80.51 $\pm$ 3.58	63.10 $\pm$ 3.02	61.95 $\pm$ 2.73	48.19 $\pm$ 2.96	50.41 $\pm$ 3.51
Pre-train	DGI	78.82 $\pm$ 4.75	79.44 $\pm$ 2.97	60.69 $\pm$ 3.62	58.75 $\pm$ 3.98	47.93 $\pm$ 4.39	49.34 $\pm$ 3.11
	GraphCL	78.22 $\pm$ 6.88	79.32 $\pm$ 4.96	62.21 $\pm$ 4.04	60.07 $\pm$ 4.38	47.66 $\pm$ 3.19	50.92 $\pm$ 2.87
	SimGRACE	79.59 $\pm$ 5.43	80.94 $\pm$ 3.77	61.84 $\pm$ 1.30	60.31 $\pm$ 2.51	48.07 $\pm$ 3.76	51.24 $\pm$ 3.39
Prompt	GPPT	76.28 $\pm$ 4.64	77.97 $\pm$ 3.08	-	-	45.63 $\pm$ 2.65	47.38 $\pm$ 2.92
	GraphPrompt	78.34 $\pm$ 5.92	79.15 $\pm$ 2.87	-	-	48.15 $\pm$ 3.84	49.72 $\pm$ 4.58
	GraphPrompt+	79.87 $\pm$ 4.51	80.14 $\pm$ 2.12	-	-	48.38 $\pm$ 3.21	50.12 $\pm$ 3.65
	ProG	79.98 $\pm$ 5.48	80.79 $\pm$ 2.53	62.92 $\pm$ 2.19	60.30 $\pm$ 3.44	49.72 $\pm$ 3.58	51.57 $\pm$ 4.33
	FGMPL	84.53 $\pm$ 4.04	83.39 $\pm$ 1.52	66.19 $\pm$ 3.14	64.02 $\pm$ 2.29	53.99 $\pm$ 3.07	56.14 $\pm$ 3.60
IMP		4.55%	2.60%	3.27%	3.42%	4.27%	4.57%

The best results are highlighted in bold.

Table 4. Single-task Performance Comparison (%)

Corporate performance			Corporate relationship			News classification		
ACC			ACC			ACC		
F1			F1			F1		
HAN	53.31 $\pm$ 2.17	54.56 $\pm$ 2.64	N2V	61.86 $\pm$ 3.32	60.01 $\pm$ 2.55	EventX	45.65 $\pm$ 2.53	47.36 $\pm$ 3.12
HATS	77.93 $\pm$ 3.00	79.47 $\pm$ 2.81	VGAE	64.20 $\pm$ 2.93	62.51 $\pm$ 3.22	SCStory	50.18 $\pm$ 2.76	51.98 $\pm$ 3.01
FGMPL	84.53 $\pm$ 4.04	83.39 $\pm$ 1.52	FGMPL	66.19 $\pm$ 3.14	64.02 $\pm$ 2.29	FGMPL	53.99 $\pm$ 3.07	56.14 $\pm$ 3.60

The best results are highlighted in bold.

method continued to outperform all baselines across the three tasks, with improvements of 4.55%, 3.27%, and 4.27%, respectively.

A comparative analysis of the results in Tables 2 and 3 reveals several noteworthy observations. Firstly, we noted that in the full dataset scenario, the performance disparities among different methods were significantly reduced compared to the few-shot setting. This convergence suggests that when data is abundant, even relatively simple methods can learn effective feature representations. Nevertheless, prompt-based methods still maintained superiority over supervised and pre-trained methods across all tasks, albeit with a diminished margin of advantage. Notably, the FGMPL method retained its superior performance, achieving the highest accuracy and F1 scores across all tasks. In the full dataset environment, pre-trained methods continued to outperform supervised learning approaches, though the margin of improvement was less pronounced than in the few-shot setting. This observation indicates that while the prior knowledge encapsulated in pre-trained methods remains beneficial in data-rich environments, its impact is relatively attenuated compared to data-scarce conditions. Within the supervised learning category, GAT and RGAT maintained a slight edge over GCN in the company relationship and news classification tasks. This persistent advantage suggests that the benefits of attention mechanisms in these specific domains are not confined to few-shot scenarios but extend to large-scale data settings as well. Finally, to assess generalization in single-task scenarios, Table 4 compares FGMPL with single-task baselines.

Table 5. Statistics of NASDAQ Dataset

Data	Node	Node type	Edge	Edge type
News graph	19,068	4	41,933	1
Corporate graph	26,451	3	87,757	5
All graph	41,284	4	124,643	6

Table 6. Performance Evaluation (%) on NASDAQ Dataset

Data	Methods	Corporate performance		Corporate relationship		News classification	
		ACC	F1	ACC	F1	ACC	F1
100-shot	RGAT	61.31 $\pm$ 10.94	60.35 $\pm$ 6.87	34.28 $\pm$ 5.83	33.96 $\pm$ 6.14	21.16 $\pm$ 4.93	21.29 $\pm$ 6.12
	GraphCL	69.47 $\pm$ 11.92	68.28 $\pm$ 7.66	36.34 $\pm$ 5.64	36.52 $\pm$ 7.26	24.89 $\pm$ 4.62	23.38 $\pm$ 3.71
	ProG	70.35 $\pm$ 11.68	70.94 $\pm$ 6.49	42.82 $\pm$ 9.32	41.28 $\pm$ 11.66	28.47 $\pm$ 6.20	30.25 $\pm$ 6.87
	FGMPL	73.86 $\pm$ 9.52	72.68 $\pm$ 7.35	44.74 $\pm$ 7.86	43.75 $\pm$ 8.64	31.15 $\pm$ 6.94	31.53 $\pm$ 5.85
All data	RGAT	78.93 $\pm$ 4.68	78.40 $\pm$ 3.92	62.35 $\pm$ 3.26	61.06 $\pm$ 2.94	45.78 $\pm$ 3.24	46.33 $\pm$ 3.88
	GraphCL	78.66 $\pm$ 7.12	78.23 $\pm$ 5.38	62.37 $\pm$ 4.31	61.42 $\pm$ 4.76	47.90 $\pm$ 3.68	48.60 $\pm$ 4.61
	ProG	78.08 $\pm$ 5.01	78.81 $\pm$ 4.93	62.94 $\pm$ 2.47	62.11 $\pm$ 3.83	49.07 $\pm$ 4.18	49.41 $\pm$ 2.80
	FGMPL	79.22 $\pm$ 4.10	80.20 $\pm$ 2.28	63.65 $\pm$ 2.15	62.93 $\pm$ 2.56	50.51 $\pm$ 3.22	51.34 $\pm$ 3.04

The best results are highlighted in bold.

5.3 Cross-market Dataset Extension and Generalizability Evaluation

To assess cross-market generalizability beyond the Chinese financial market, we construct an additional NASDAQ dataset (Q2 2016) by integrating five external sources: NYT Articles,⁸ Wikinews Article,⁹ FinSen,¹⁰ American News,¹¹ and Relato Business Graph.¹² As shown in Table 5, notably, the corporate relationship graph in the NASDAQ dataset includes only three relation types—investment, supply, and group—which form a subset of the relation schema used in our Chinese dataset. This choice is due to data availability and enables a controlled cross-market evaluation under a partially shared relational ontology.

Since U.S. news is in English while our text encoder is pre-trained on large-scale Chinese financial corpora, we automatically translate English news into Chinese during pre-processing. After translation, we apply the same pre-processing pipeline as the Chinese dataset and reuse the same text encoder and GNN backbone. This design focuses the evaluation on the transferability of graph structure modeling and the prompting mechanism rather than retraining an English-oriented encoder.

We evaluate a realistic rapid adaptation scenario: we keep the model pre-trained on the Chinese market, freeze the backbone, and only train the prompt parameters on the NASDAQ dataset, simulating low-cost adaptation to a new market. Results in Table 6 show that FGMPL consistently outperforms RGAT, GraphCL, and ProG under both 100-shot and full-data settings across corporate performance prediction, corporate relationship classification, and news classification, demonstrating robust cross-market adaptation.

⁸<https://www.kaggle.com/datasets/aryansingh0909/nyt-articles-21m-2000-present>.

⁹<https://www.kaggle.com/datasets/datagator/wikinews-article-dataset>.

¹⁰<https://www.kaggle.com/datasets/eaglewhl/finsen-financial-sentiment-dataset>.

¹¹<https://www.kaggle.com/datasets/jonasbecker98/news-website-articles>.

¹²<https://www.kaggle.com/datasets/thedevastator/relato-business-network-graph-373663-domain-conn?select=links.csv>.

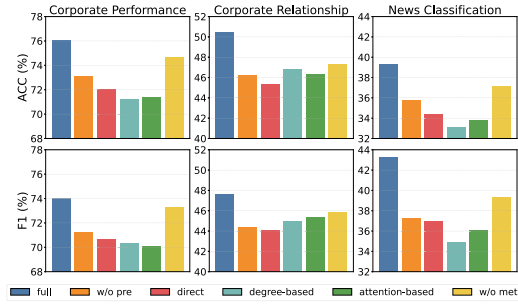


Fig. 5. Ablation study.

5.4 Ablation Study

To comprehensively evaluate the effectiveness of our proposed method, we conduct a series of ablation experiments. We focus on three key components: single-view pre-training, expert mixing mechanism, and meta-learning. The results of these experiments are presented in Figure 5. (1) Single-view Pre-training: We employ the pre-training approach from GraphCL as our baseline, followed by our subsequent components. This configuration allows us to assess the impact of multi-view learning compared to single-view learning in our framework. (2) MoE Mechanism: We compare our MoE mechanism against several alternatives: Direct Weighting, Attention-based Weighting and Degree-based Weighting. These comparisons aim to demonstrate the superiority of our proposed expert mixing mechanism in capturing and integrating diverse graph characteristics. (3) Meta-learning: We evaluate the effectiveness of our meta-learning approach by comparing it with a variant without the meta-learning step, denoted as “w/o meta.”

Through a comparative analysis of the complete model and its single-view pre-trained variants, we observe that the designed multi-view pre-training is crucial for comprehensively capturing heterogeneous graph information, thereby establishing a robust foundation for downstream tasks. Regarding the MoE mechanism, our proposed MoE method demonstrates excellent performance in effectively capturing and integrating diverse graph features. Finally, by contrasting models with and without the meta-learning step, we find that meta-learning contributes significantly to model performance.

5.5 Efficiency Analysis

Figure 6(a) illustrates the impact of varying token quantities on model performance. The results demonstrate that our model achieves excellent performance with a significantly reduced number of tokens, highlighting the efficiency of our model design. In comparison to ProG, which utilizes 10 tokens, our approach maintains or enhances performance while substantially decreasing the token count.

Building upon the token quantity experiments, we further investigate the activation ratio in the MoE mechanism. Figure 6(b) presents comprehensive experimental results. The results reveal that most tasks achieve optimal performance at relatively low activation ratios (around 0.2), with performance generally showing a declining trend as the ratio increases. This phenomenon suggests that moderate expert participation is more conducive to effective feature extraction from complex financial data, while excessive activation may lead to information redundancy. Full data training consistently demonstrates superior and more stable performance compared to 100-shot scenarios, underlining the importance of data scale in model effectiveness. The corporate relationship prediction task exhibits unique characteristics, showing stable positive responses

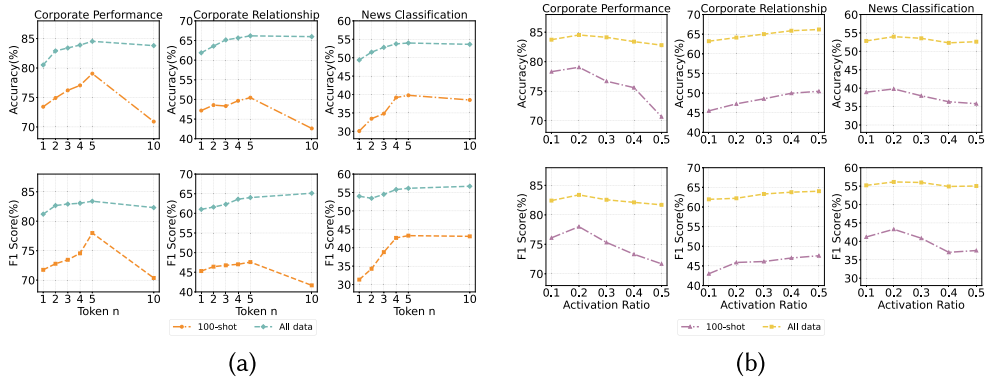


Fig. 6. (a) Impact of token numbers. (b) Impact of top-k experts.

Table 7. Performance Evaluation (%) on Different Settings

Tasks	3-way 20-shot		3-way 50-shot		5-way 20-shot		5-way 50-shot		3-way 100-shot	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
Corp. performance	76.81	75.16	78.21	77.12	75.30	73.97	77.90	76.85	79.05	77.99
Corp. relationship	47.83	45.70	49.41	46.76	47.10	45.13	48.87	46.01	50.46	47.59
News classification	37.24	39.70	38.60	42.51	36.72	39.34	37.85	40.06	39.79	43.28

to higher activation ratios, which may be attributed to the inherent complexity of corporate relationships requiring more collaborative expert involvement.

To thoroughly evaluate the model's performance in few-shot scenarios, we focused our analysis on experimental configurations with 20-shot and 50-shot settings. As shown in Table 7, our proposed framework demonstrates robust capability in handling challenging multi-class tasks even with extremely limited samples. When comparing 3-way and 5-way configurations, we observed that although the 5-way setting presents a more challenging classification scenario, the model exhibits only marginal performance degradation. This finding substantiates the scalability of our graph prompt learning framework in effectively managing classification tasks with varying numbers of categories. Through the integration of meta-learning techniques, the model demonstrates remarkable rapid adaptation capabilities. Even under limited sample conditions (e.g., 20-shot configuration), the model maintains stable performance, validating the significant advantages of our framework in addressing few-shot learning scenarios commonly encountered in the financial domain.

For the time complexity, we compare FGMPL with ProG [68], our strongest prompt-based baseline. ProG augments the input graph with a small prompt graph and keeps the time complexity linear in the number of nodes and edges:

$$O(L(N + n)d^2 + L(M + m)d + (N + n)d), \quad (31)$$

where N, M are the numbers of original nodes and edges, and $n \ll N, m \ll M$ come from the prompt graph. Our framework adopts the same prompting paradigm; hence, the encoder part has the same asymptotic complexity as ProG. The additional cost in FGMPL mainly comes from two components: (i) a sparse Financial Multi-Experts Prompting (MoE) module, and (ii) multi-view contrastive pre-training.

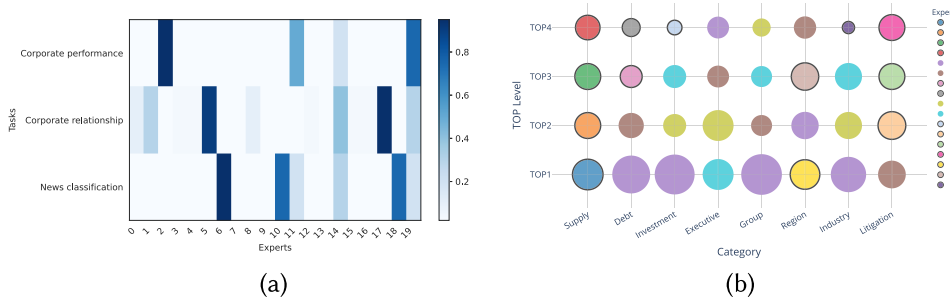


Fig. 7. (a) Visualization of the frequency that experts are selected for each task. (b) Visualization of the frequency that experts are selected for corporate relationship tasks.

Let $N' = N + n$ denote the number of nodes after prompting. For each node, the MoE layer only activates a small subset k out of R experts, leading to an extra cost:

$$O(N'(Rd + kd^2)), \quad (32)$$

where Rd accounts for computing routing/gating scores over R experts and kd^2 accounts for applying the k selected expert transformations. In our implementation, R and k are small constants, so this term remains linear in N' and amounts to a modest constant-factor overhead over ProG during inference. The multi-view contrastive objective introduces several GNN forward-backward passes (for a fixed number of views) and a batch-wise similarity computation:

$$O(B^2d). \quad (33)$$

Importantly, this is performed once offline in the pre-training stage and does not affect online prediction latency.

Overall, FGMPL retains linear-time inference with respect to the graph size, while trading a slightly higher offline training cost for improved performance and adaptability in financial scenarios.

5.6 Interpretability Analysis

To make the model behavior more transparent, we provide a systematic interpretability analysis from two complementary perspectives: (i) task-level expert routing patterns and (ii) representation geometry of the learned embeddings. We first investigate how the MoE prompting layer routes different tasks to different experts. Specifically, for each downstream task, we compute the average expert activation (i.e., routing weight) over the corresponding evaluation set and visualize the results as a heatmap in Figure 7(a). The observed routing patterns are sparse and clearly task-dependent, showing that each task consistently relies on a small subset of experts rather than uniformly activating all components. Concretely, corporate performance prediction tends to activate experts associated with price/fundamental signals, corporate relationship prediction emphasizes experts capturing graph-structural and inter-firm interaction patterns, and news classification mainly relies on experts focusing on textual semantics. This task-level specialization indicates that the MoE layer learns a meaningful decomposition of heterogeneous financial information, improving interpretability by making “which experts matter for which task” explicit.

We further drill down into the corporate relationship task and analyze expert routing across different relation types, summarized in Figure 7(b). The size of the bubbles in the figure represents the relative importance of each expert within a category, reflecting the normalized contribution of their activation values (the proportion of the four activation values). Experts outlined in black are exclusively dedicated to specific relationship types. For instance, the “Supply” relationship

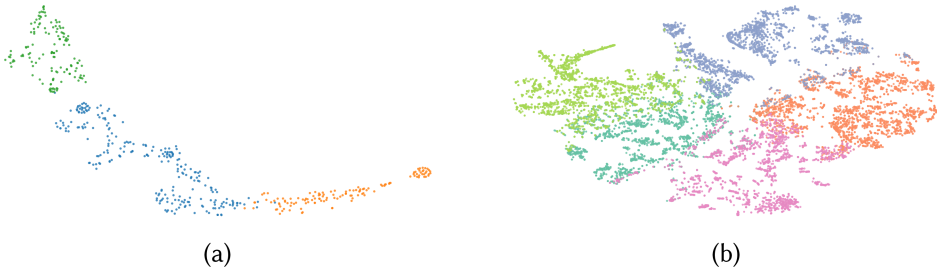


Fig. 8. (a) Visualization of the corporate performance embeddings. (b) Visualization of the corporate relationship embeddings.

exclusively activates specialized experts, reflecting its unique characteristics and low correlation with other types. In contrast, the “Executive” and “Group” relationships rely solely on shared experts, consistent with their high-frequency samples shown in Figure 2(b). This indicates that the mechanism tends to leverage shared experts to handle high-frequency relationship types, thereby optimizing resource utilization. The “Debt” and “Litigation” relationships, however, exhibit a hybrid pattern, activating both shared and exclusive experts. Shared experts (e.g., expert 5) capture cross-relationship features, while exclusive experts distinguish unique characteristics, effectively addressing relationship diversity. This balanced allocation of exclusive and shared experts enables the model to combine specificity and generalization, offering a robust approach for handling multi-relationship tasks, especially in imbalanced data scenarios.

Finally, to understand what FGMPPL learns at the representation level, we apply t-SNE to the learned graph-level embeddings and visualize them in Figure 8(a) and (b). Each color represents a unique class label. The resulting low-dimensional layouts form well-structured clusters that align with the ground-truth labels and their semantic affinities, suggesting that FGMPPL organizes companies/news into a representation space that preserves meaningful financial structure.

5.7 Case Study

The first case (Figure 9) powerfully demonstrates the exceptional performance of our proposed FGMPPL. Specifically, the model successfully predicted that the company’s stock performance would outperform the market by more than 5% (with an actual result of 9.81%). In the analysis of corporate relationship networks, the model accurately identified multiple investment relationships, demonstrating a profound understanding of complex corporate interactions. Notably, the model misclassified the relationship with 000018.SZ as a debt relationship, when it was actually a loan contract dispute. While not entirely accurate, this misclassification reflects the model’s ability to capture the essence of financial disputes, as loan contract disputes are often closely related to debt issues, sharing many characteristics such as involving creditors, debtors, and repayment obligations. This subtle distinction highlights the complexity of financial relationships while also indicating areas for improvement in the model’s handling of nuanced semantic differences. In the task of news event classification, the model accurately categorized relevant news under “Measures for the real estate industry” rather than simplistically classifying it under banking-related topics.

As illustrated in Figure 10, the news event regarding the strategic partnership between Huawei and **Jianghuai Automobile (JAC)** effectively demonstrates the practical value of our framework. Upon receiving new news data, the system only needs to extract entity nodes (such as Huawei and JAC) and integrate them into the existing corporate relationship network. Through the pre-trained multi-view contrastive learning model, the system efficiently obtains representations of these newly added nodes, organically integrating news text information with the structural information of the

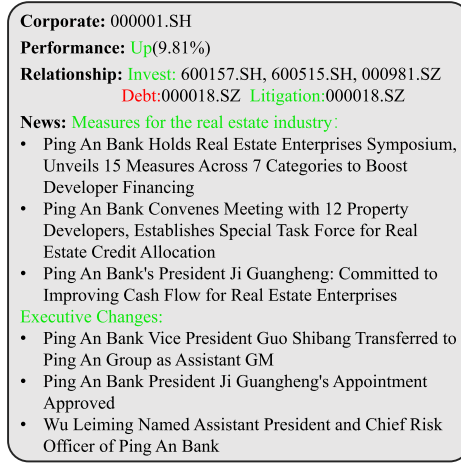


Fig. 9. Case study 1. Multi-task prediction results.

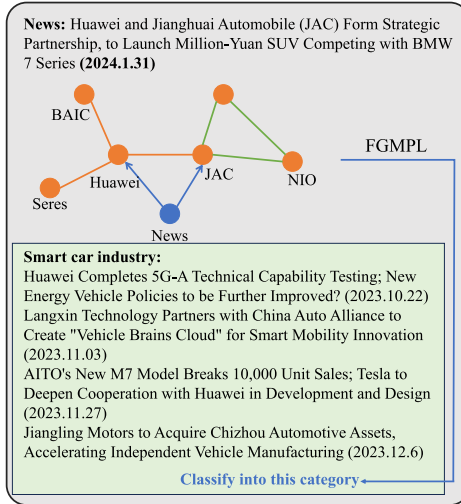


Fig. 10. Case study 2. Emerging news category results.

corporate relationship network. Our framework eliminates the need for retraining the entire model, instead utilizing the well-trained FGMPL to directly classify news events. The case study reveals that the model successfully captures Huawei's strategic positioning in the automotive industry and its collaborative network with multiple automotive manufacturers, accurately reflecting the integration trends within China's new energy vehicle industry chain. These efficient and flexible characteristics make our framework particularly suitable for analyzing the evolution of corporate relationships and industry dynamics in rapidly developing emerging sectors.

6 Conclusion

In this article, we propose FGMPL, a novel unified framework based on graph prompt learning for comprehensive corporate profiling. By unifying tasks into a universal graph-level format and employing multi-view contrastive learning, our framework successfully captures complex

interactions between corporations. The proposed Financial Multi-Experts Prompting mechanism, combining prompt learning with MoE, demonstrates superior capability in handling heterogeneous corporate data and adapting to newly emerging entities. Furthermore, our meta-learning enhanced prompt tuning approach enables rapid task adaptation. Experimental results show significant improvements across multiple aspects of corporate profiling, with accuracy gains of 7.57%, 6.39%, and 8.75% in few-shot scenarios for corporate performance prediction, relationship classification, and news event categorization, respectively. These results, along with comprehensive ablation studies and case analyses, validate our framework's effectiveness in constructing comprehensive corporate profiles and its practical utility in real-world applications.

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